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Rock Mechanics (MIN E – 620)
Homework No. 3



1 Problem

Dips Software

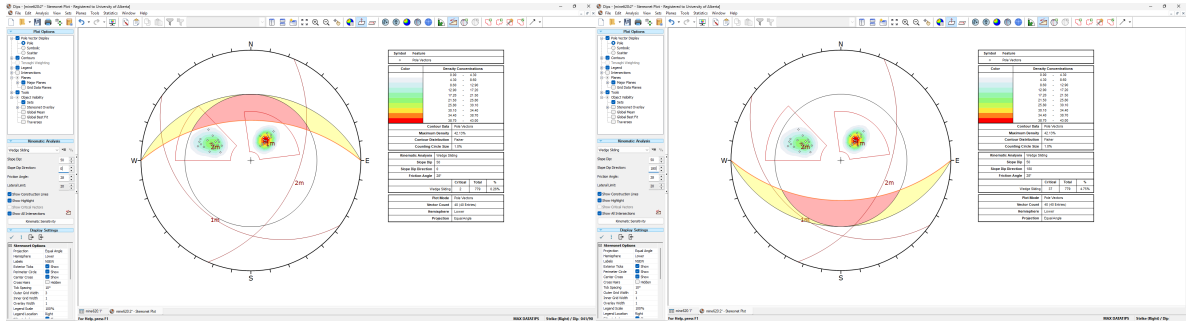
A given quarry is roughly rectangular, with the long side of the rectangular being the east sideline. The quarry has historically used a 50-degree slope angle on all sides. Typically, the quarry is dry but heavy rains have been known to saturate the area. The angle of friction of the 28 degrees. Recently, the quarry has been experiencing a series of small wedge failures. You have conducted a discontinuity survey to determine the likelihood that these small failures indicate a significant problem.

Strike / Dip

125 25 119 2 7 126 25 25 37
127 23 111 27 24 42 27 33
129 29 123 18 25 47 21 41
128 23 127 26 30 37 20 53
131 25 129 23 18 39 28 40
140 23 128 33 23 53 36 51
127 20 131 31 18 51 12 42
123 21 128 37 16 48 33 45
121 32 125 25 32 39 15 53
122 25 125 30 23 40 15 32

2 Anlysis

Input the gaven data to Dips, we can get the results by kinematic analysis.



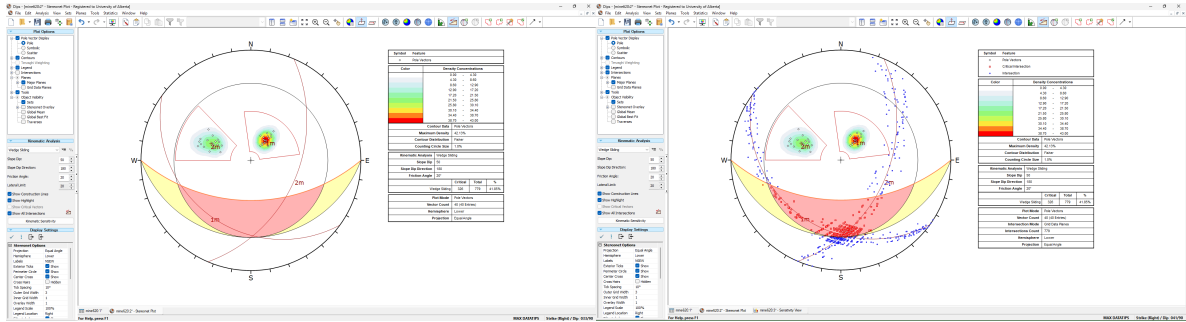
(a) The results of slope dip direction of 0

(b) The results of slope dip direction of 180

Figure 1: The current situation of slope dip direction of 0 and 180

Based on the current results, for slope dip direction of 0° (as shown in the Figure 1a), the kinematic analysis indicates low risk of wedge sliding, critical percentage is 0.26%, the joint orientation are outside the the critical sliding window. However, for slope dip direction of 180 (as shown in the Figure 1b), the critical sliding window is very close to the joint intersection, which indicates a potential risk of wedge sliding, the critical percentage is 4.75%.

For the slope dip direction of 180, we can get the results as shown in the Figure 2, if the friction angle decrease to 20 by heavy rains.



(a) The results of slope dip direction of 180 with 20 friction angle

(b) The results of slope dip direction of 180 with 20 friction angle with intersections

Figure 2: When the friction angle decrease to 20

As shown in the Figure 2a and Figure 2b, the critical percentage increase to 41.85%, which indicates a very high risk of wedge sliding, the joint intersection is well within the critical sliding window. Some necessary methods, such as rock bolts, should be taken to prevent the wedge sliding.

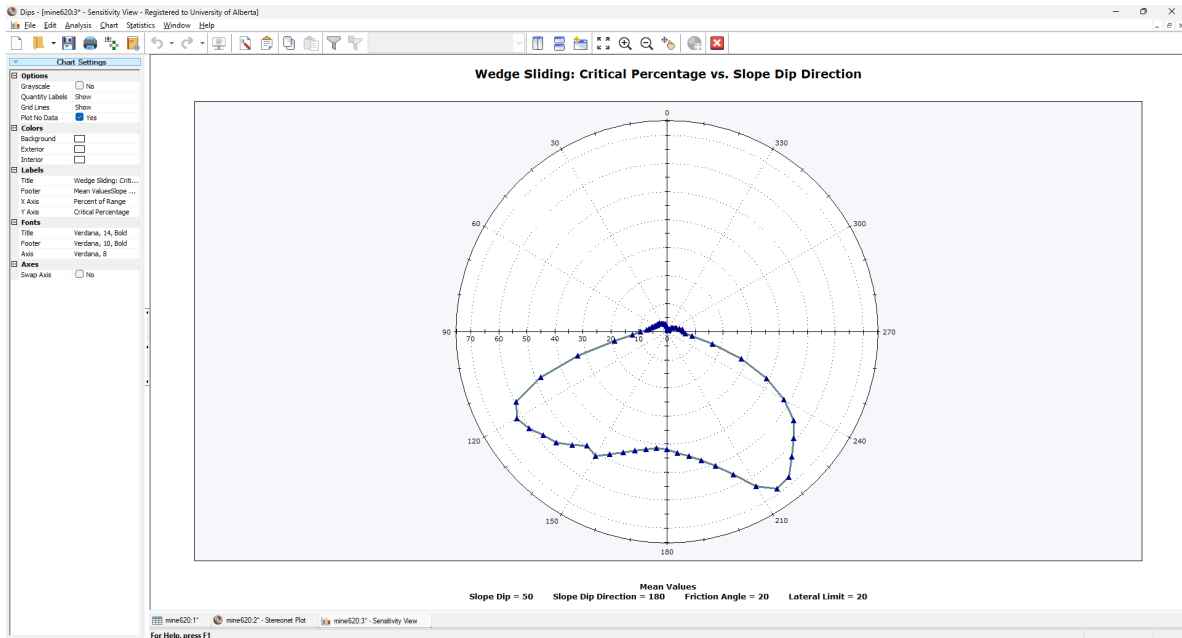


Figure 3: The radar map analysis of wedge sliding: critical percentage vs. slope dip direction

Performing the further analysis of the critical percentage with different slope dip direction, we can get the radar map as shown in the Figure 3. There are two peaks in the radar map, which are around 120 and 220. At the same time, the critical percentage is very low when the slope dip direction is in around [0-90] and [270-359].

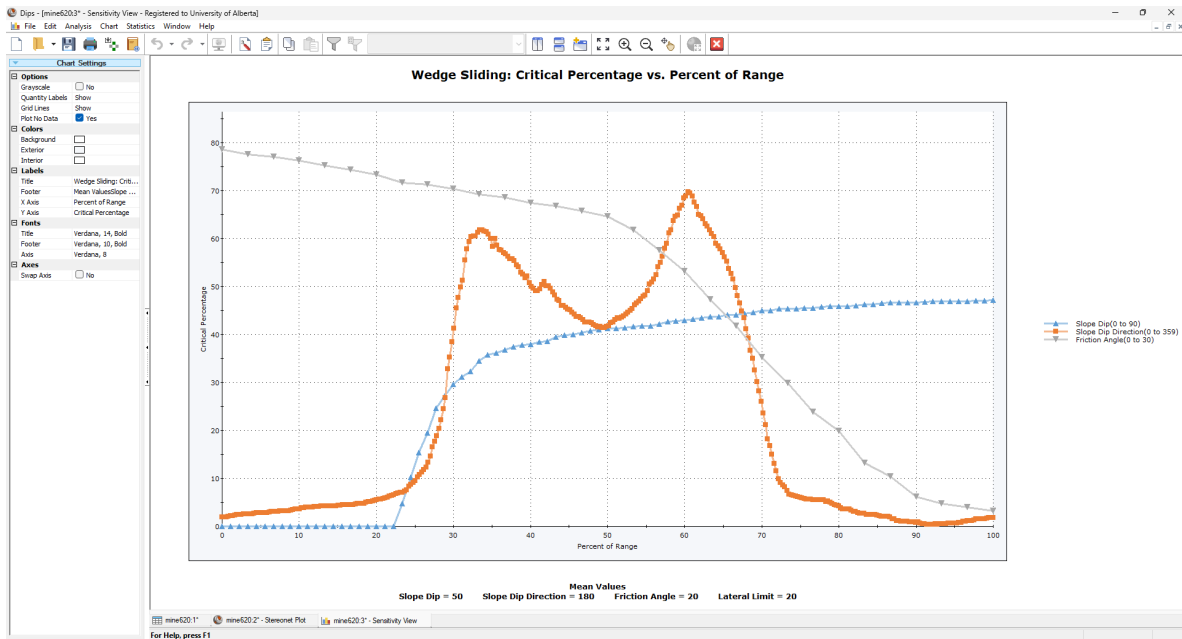


Figure 4: The critical percentage analysis of slope angle, friction angle and slope dip direction

3 Summary

Combining the analysis of slope angle, friction angle and slope dip direction, we can get the results as shown in the Figure 4. we need focus on the slope dip direction range of [110-250], which has a higher risk of wedge sliding. If the friction angle decrease or the rock severely weathered, the risk of wedge sliding will increase significantly. Should take some necessary methods, such as rock bolts or shotcrete or drainage to prevent the wedge sliding.